

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Discrete Mathematics for AI | Course Code: 21AI31 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. Solve the recurrence relation $a_n = 5a_{(n-1)} - 6a_{(n-2)}$ with initial conditions $a_0 = 1$, $a_1 = 4$ using the Characteristic Equation Method.

[8 Marks]